

Koji Wong

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EDUCATION

Washington University in St. Louis

Aug 2024 – May 2028

B.S. Computer Science + Mathematics — Biomedical Data Science Minor — GPA: 3.88

St. Louis, MO

- **Courses:** Data Structures & Algorithms, Linear Algebra, Real Analysis, Biomedical Computing, Probability & Stats for Engineering, Machine Learning & Pattern Classification, Object-Oriented Programming
- **Awards:** IMC Launchpad SWE (May 2026), 5x Hackathon Winner (**\$11,850**), Morgan Stanley Tokyo Early Insights Fellow

EXPERIENCE

Incoming Software Engineer Intern

Summer 2026

Capital One

Richmond, VA

Software Engineer Fellow

Jan 2026 – Jan 2026

Jane Street

New York City, NY

- Learning about applications of math, data science, and computer science to quantitative trading & development

Machine Learning Research Assistant

Dec 2025 – Present

Washington University in St. Louis

St. Louis, MO

- Developing analysis infrastructure for observing role of large language models and information accessibility in politics using R and Pytorch. Supervised by Dr. Jacob Montgomery (head of political data science)

Software & Machine Learning Engineer Intern

Jul 2025 – Aug 2025

Code Four (YC X25)

San Francisco, CA

- Sole developer of a **fullstack video redaction** web app tool used by **100+ police departments** using Next.js, TypeScript, TailwindCSS, with computer vision pipelines and video streaming architecture deployed in GCE, GCS, **NoSQL Firestore**
- Created computer vision pipeline using **Python, YOLOv11 & BotSort** with optimized post-processing using **Facenet & OCR** to efficiently identify key objects in police bodycam footage, saving police officers thousands of hours
- Built scalable **GCP** microservices architecture for video streaming & computer vision tasks using **Kubernetes**

Software Engineer Intern

May 2025 – Aug 2025

New York City Mayor's Office for Economic Opportunity

Remote

- Replatformed legacy eligibility logic into **Python AWS Lambda** functions and deployed via **AWS API Gateway** using infrastructure-as-code, improving performance and maintainability by 70% for **200,000 New Yorkers**
- Built **CI/CD** pipelines with version control & wrote integration & load tests to support **200,000+** monthly API requests.
- Collaborated with civic technologists using **Agile** methodologies to ensure compliance with data privacy requirements.

Artificial Intelligence Research Assistant

Nov 2024 – May 2025

Washington University in St. Louis

St. Louis, MO

- Built large-scale **Habitat simulator** pipelines to collect training data for autonomous vehicle decision systems.
- Customized simulation configs and agent logic in **PyTorch** to improve realism and coverage. Supervised by Dr. Sibai

PROJECTS

Generalized Cameraman AI Agent | *Python, SciPy, OpenCV, React, Electron*

May 2025 – May 2025

- Created AI cameraman desktop app with advanced **algorithm development** using **Kalman filters, spline interpolation, multimodal models**, & **OpenCV** to detect and smoothly zoom/pan to follow core action in video input.
- Built cross-platform GUI with **React** and **Electron**, supporting real-time preview and parameter tuning.

Official Supabase Community Starter Template | *React, TypeScript, Supabase*

May 2025-May 2025

- Created modern **mobile development** social media starter with design patterns using **React, Zustand, TailwindCSS**, and Supabase auth/storage APIs.
- Accepted to **official Supabase community templates**, supporting indie developers worldwide. **1000+** total cloners

Winning Hackathon Projects | *NextJS, FastAPI, Python, JavaScript, React, Swift*

Oct 2024 – Jan 2026

- **Google DevFest ElevenLabs Track ** XR voice assistant for EMTs. Writes ePCRs via voice video, voice commands, pulls vital patient data (allergies etc) using face rec, and browser use integrations for hospital form-filling
- **YC Fullstack Hackathon ** Developed AI agents to 10x case worker workflows: call agents, RAG, and email agents
- **Unveil AI ** (Finalist @ WashU Google DevFest): Built Gemini-powered onboarding tool with **NextJS** frontend, **FastAPI** backend, and **AI software engineer** call-agent with vectorized codebase database for knowledge retrieval
- **InternX ** (2nd @ WashU Meta × 8VC): Created **AI-powered** networking tool using **React & Express.js**.
- **Bear Eats ** (Winner @ Hack WashU): Built Swift iOS **mobile app** to recommend nutritious campus meals

TECHNICAL SKILLS

Languages: Python, Java, C, C++, C#, Swift, TypeScript/JavaScript, HTML, CSS, SQL, Zsh

Technologies: PydanticAI, LangGraph, React, Next.js, Expo, Express, FastAPI, PyTorch, TensorFlow, Tailwind, Pandas, PostgreSQL, Git, Supabase, Google Cloud Platform, Docker, FastMCP, Postman, Zustand, OpenCV, Electron

Practices: CI/CD, Microservices, NoSQL Databases, Algorithm Development, Data Structures